

MATLAB: Session 1

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What is MATLAB?

MATLAB is a programming and numerical computing platform used by mathematicians, engineers, and scientists to analyze data, develop algorithms, and create models. In this course, we will use MATLAB to manipulate objects from Geometry and Linear Algebra, such as vectors and matrices.

To use MATLAB, you will need a MathWorks account and a valid MATLAB license. The software is provided for the entire duration of your studies. You can activate your license by visiting this webpage and logging in with your *@studenti.unitn.it* account. After activation, you can either use MATLAB Online through this website or download it to your computer. If you prefer to use the university's computers, simply log in with your *@studenti.unitn.it* credentials, and you will find a ready-to-use MATLAB installation on the desktop.

How to use these sheets?

The exercise sheets will be uploaded to the Moodle page of the course a couple of days before each tutoring session. Try these exercises *before* the session, so you can arrive with concrete questions. During the session, we will solve and discuss them together. Afterwards, similar problems will appear on MATLAB Grader for self-practice.

MATLAB Grader is a browser-based tool from MathWorks that lets us create interactive MATLAB coding problems and automatically grade your submissions, with instant feedback and integration with common Learning Management Systems (LMS). You can access MATLAB Grader through this website. To submit your solutions, you must be enrolled in our MATLAB Grader course. Please email *vincenzo.isoldi@studenti.unitn.it* or *francesco.devitis@studenti.unitn.it* from your *@studenti.unitn.it* account to be added.

Exercises.

Part A – First Steps

- A.1** Use MATLAB as a calculator: compute $23 + 31$, $5.75 \times \frac{7}{2}$, $\frac{10}{3} \times \frac{23}{2}$, and $\sqrt{27}$.
- A.2** Define two variables $\mathbf{a} = 12$ and $\mathbf{b} = -7 + \frac{2}{31}$. What happens if you add a semicolon ; at the end of the definition?
- A.3** Compute their sum and store it in a new variable $\mathbf{c}$.
- A.4** Clear all the variables stored, using `clear all`.
- A.5** Add “MATLAB: Session 1” as a comment, using %.

Part B – Vectors

- B.1** Create the vectors

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{w} = [3 \ 3 \ 3 \ 3].$$

Assign them to the variables $\mathbf{v}$ and $\mathbf{w}$.

- B.2** Compute $\mathbf{v} + \mathbf{w}^t$ and assign the result to a variable $\mathbf{t}$. Does it make sense to compute $\mathbf{v} + \mathbf{w}$?
- B.3** Access the second entry of $\mathbf{v}$ and assign it to the variable $\mathbf{v2}$.
- B.4** Replace the third entry of $\mathbf{v}$ with the number 10.

B.5 Check whether $\mathbf{v}$ is equal to $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$.

B.6 Check whether $\mathbf{v}$ is equal to $\begin{bmatrix} 1 \\ 2 \\ 10 \\ 4 \end{bmatrix}$.

Part C – Matrices

C.1 Create the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 1 & 2 & 0 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 3 & 1 \end{bmatrix}.$$

Assign it to the variable **A**.

C.2 Find the entry of A in row 2, column 3 and assign it to **a23**.

C.3 Extract the first row of A and assign it to **row1**.

C.4 Extract the second column of A and assign it to **col2**.

C.5 Create the identity matrix of size 3×3 . Assign it to **I3**.

C.6 Create a zero matrix of size 2×4 . Assign it to **Z24**.

C.7 Create a new matrix B obtained by adding 2 times the identity matrix to A :

$$B = A + 2I_4.$$

Assign it to **B**.

C.8 Create a new matrix C by attaching to A the row $[2 \ 4 \ 7 \ 7]$ as fifth row. Assign it to **C**.

C.9 Create a new matrix D by attaching to A the column $\begin{bmatrix} 10 \\ 9 \\ 2 \\ 2 \end{bmatrix}$ as third column. Assign it to **D**.

C.10 Create a new matrix E by suppressing from A the second row. Assign it to **E**.

C.11 Compute the size of A, B, C, D, E .

Part D – Span and Membership

D.1 Create the vectors

$$\mathbf{r} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{s} = \begin{bmatrix} -2 \\ 4 \\ -6 \end{bmatrix}.$$

Assign them to the variables **r** and **s**.

D.2 Check whether $\mathbf{s} \in \text{span}\{\mathbf{r}\}$. Verify in MATLAB, by comparing componentwise ratios.

D.3 Create the vectors

$$\mathbf{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Assign them to the variables $\mathbf{p}$, $\mathbf{q}$, $\mathbf{x}$ and $\mathbf{y}$.

D.4 Find real scalars a, b such that $\mathbf{x} = a\mathbf{p} + b\mathbf{q}$, then verify in MATLAB.

D.5 Check whether $\mathbf{y}$ belongs to $\text{span}\{\mathbf{p}\}$ and to $\text{span}\{\mathbf{q}\}$, then verify in MATLAB.

Part E – Linear independence

E.1 Write a small MATLAB functions that takes as input two vectors (as columns) and checks if they are linearly independent, handling possible zeros safely.

Hint: Start by checking whether the two vectors have the same length, i.e. whether they both belong to the same $\mathbb{R}^n$. It could be useful to follow the MATLAB Onramp course to get familiar with functions.

E.2 Explain why any set of four vectors in $\mathbb{R}^3$ is linearly dependent. Provide a concrete example of four such vectors and exhibit a non-trivial linear relation among them; verify it in MATLAB.

E.3 Create the vectors

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

Assign them to the variables $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$.

E.4 Show that $\mathbf{u}_3 = \frac{3}{2}\mathbf{u}_1 + \mathbf{u}_2$. Conclude that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly dependent, while $\{\mathbf{u}_1, \mathbf{u}_2\}$ is independent. Verify in MATLAB, using the function created in **E.1**.

E.5 Create the vectors

$$\mathbf{s}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{s}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{s}_4 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}.$$

Assign them to the variables $\mathbf{s}_1$, $\mathbf{s}_2$, $\mathbf{s}_3$ and $\mathbf{s}_4$.

E.6 Reduce $\{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \mathbf{s}_4\}$ to a smaller generating set that is independent. State the dimension of the span and one possible basis; confirm relations in MATLAB, using the function created in **E.1**.